

Module 7: Logarithms – Prequiz

Name:

Date:

Four short questions. Give exact answers when reasonable and decimal approximations when requested.

1. Rewrite $3^4 = 81$ in logarithmic form.
2. Evaluate $\log_2(32)$ and $\ln(e^3)$.
3. Expand $\log\left(\frac{x^2}{y}\right)$ using logarithm laws.
4. Solve $5^{x-1} = 12$. Give an exact answer using logarithms and a decimal approximation to three places.

Module 7: Logarithms – Weekly Check-In

Name:

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Three questions. Give exact answers when reasonable and decimal approximations when requested.

1. Exponential and logarithmic forms.

- (a) Rewrite $2^5 = 32$ in logarithmic form.
- (b) Evaluate $\log_3(81)$ and $\ln(e^{-2})$.
- (c) Expand $\log\left(\frac{x^3\sqrt{y}}{z^2}\right)$ using logarithm laws.

2. Solving equations with logarithms.

- (a) Solve $4^{x-2} = 11$. Give an exact answer using logarithms and a decimal approximation.
- (b) Solve $\ln(2x - 1) = 3$. Give an exact answer and check the logarithm's domain.

Module 7: Logarithms – Weekly Check-In, continued

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3. A fading observatory signal.

An observatory is tracking a weak radio source. The measured signal power is modeled by

$$P(d) = 1800(0.84)^d,$$

where $P(d)$ is the signal power in arbitrary units and d is the distance from the receiver in kilometers.

- (a) What is the signal power at the receiver, when $d = 0$?

- (b) Find the signal power when the source is 5 km away. Round to the nearest whole unit.

- (c) At what distance will the signal power drop to 300 units? Set up and solve an exponential equation using logarithms. Round to three decimal places.